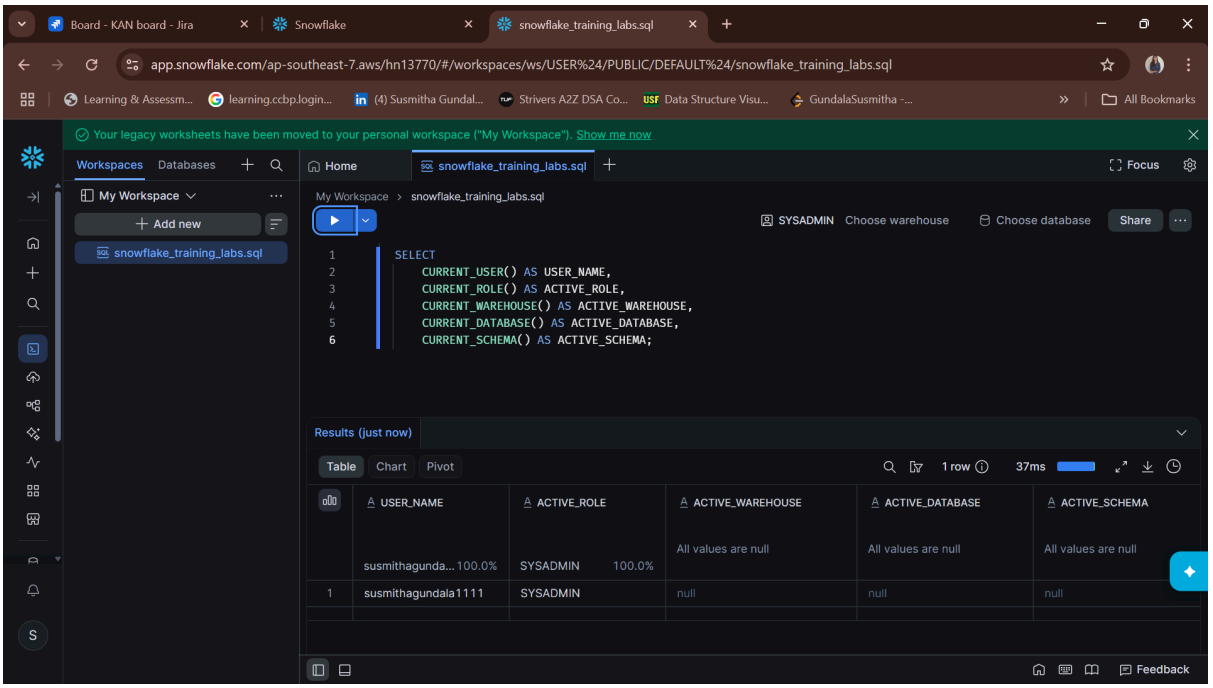
SNOWFLAKE SQL

PART 1: DATABASE AND SESSION MANAGEMENT

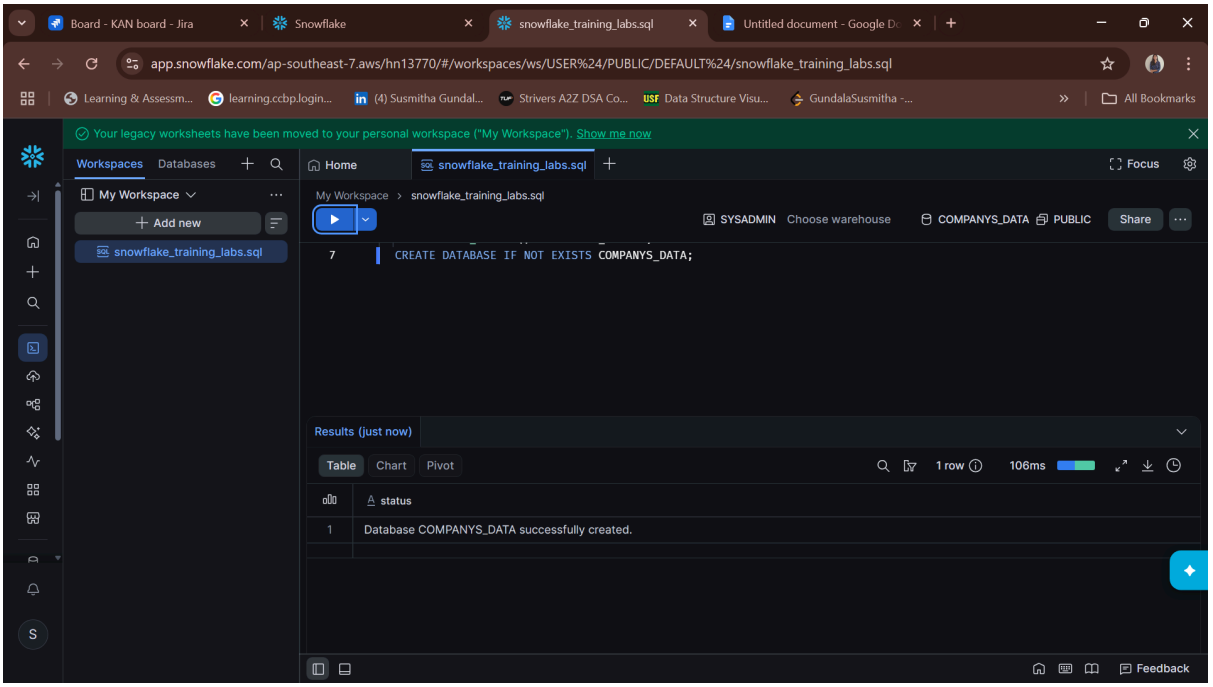
Step 1: Check Current Session Information

```sql
SELECT
    CURRENT_USER() AS USER_NAME,
    CURRENT_ROLE() AS ACTIVE_ROLE,
    CURRENT_WAREHOUSE() AS ACTIVE_WAREHOUSE,
    CURRENT_DATABASE() AS ACTIVE_DATABASE,
    CURRENT_SCHEMA() AS ACTIVE_SCHEMA;
```



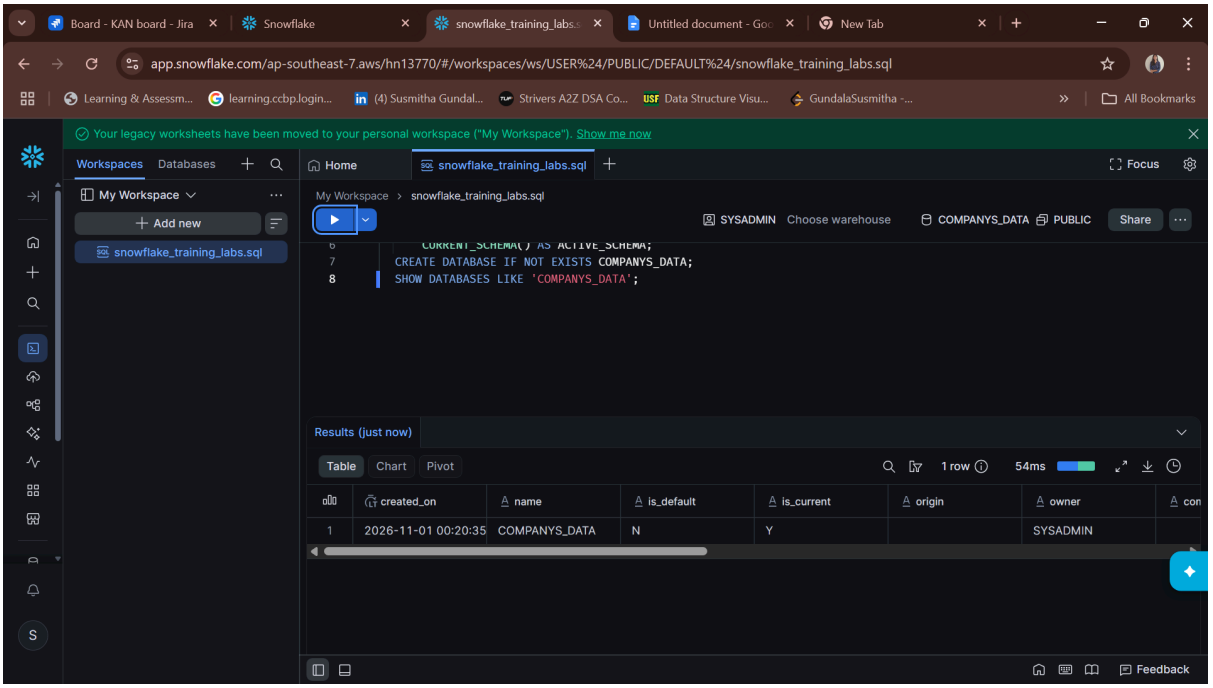
Step 2: Create a Database

```sql
CREATE DATABASE IF NOT EXISTS COMPANYS_DATA;
```



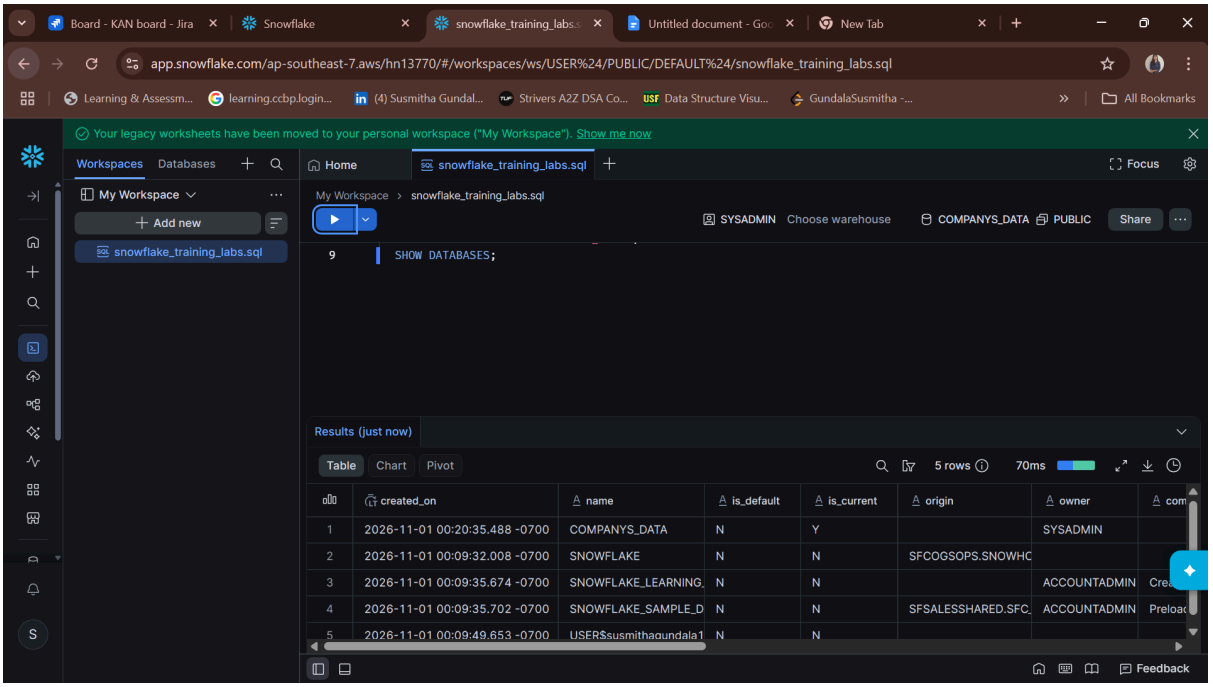
Verify the database:

```sql
SHOW DATABASES LIKE 'COMPANYS_DATA';
```



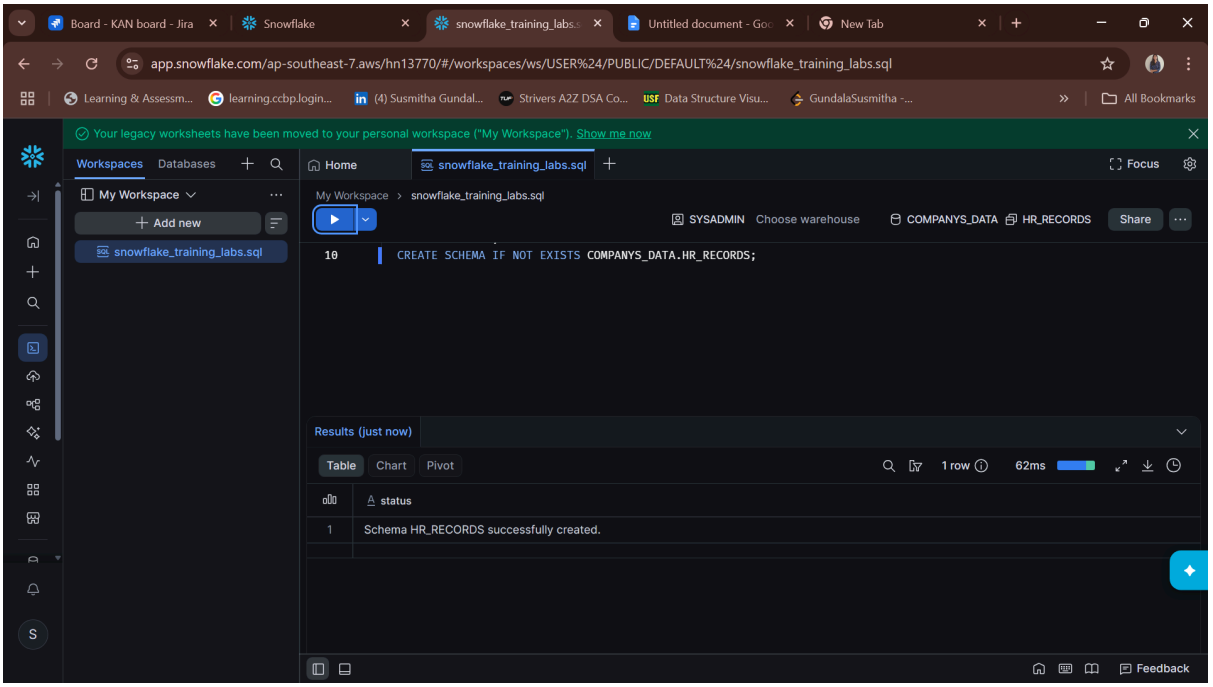
Display all databases:

```sql
SHOW DATABASES;
```



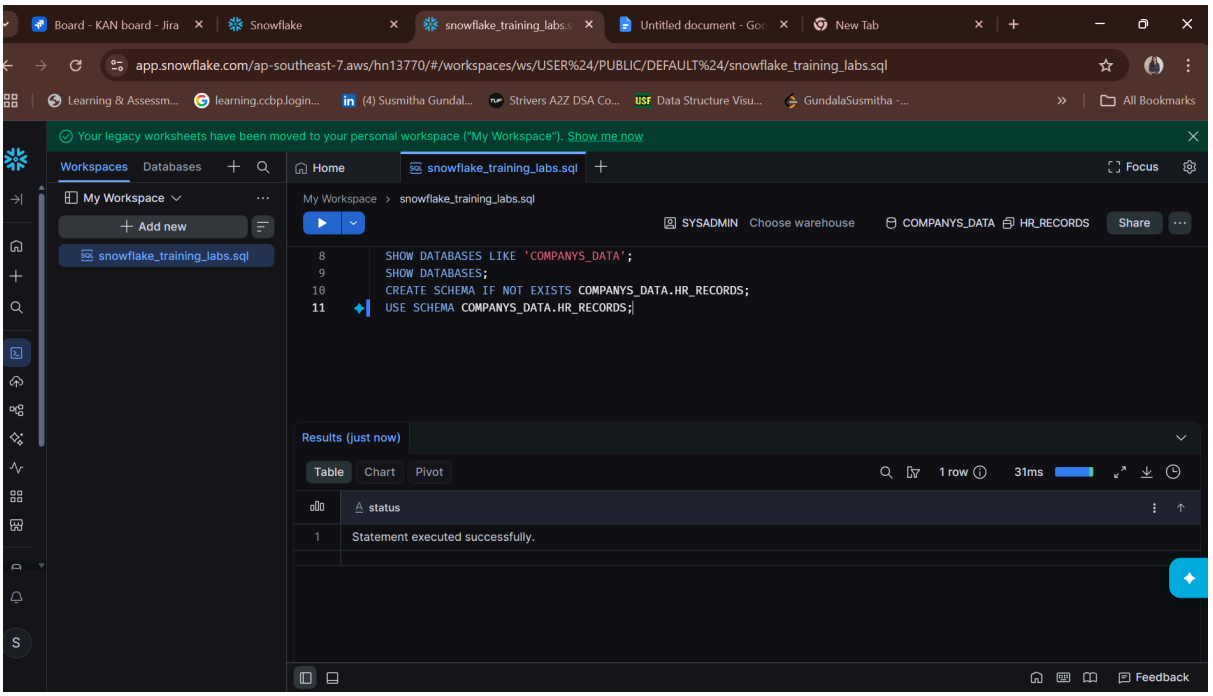
Step 3: Create a Schema

```sql
CREATE SCHEMA IF NOT EXISTS COMPANYS_DATA.HR_RECORDS;
```



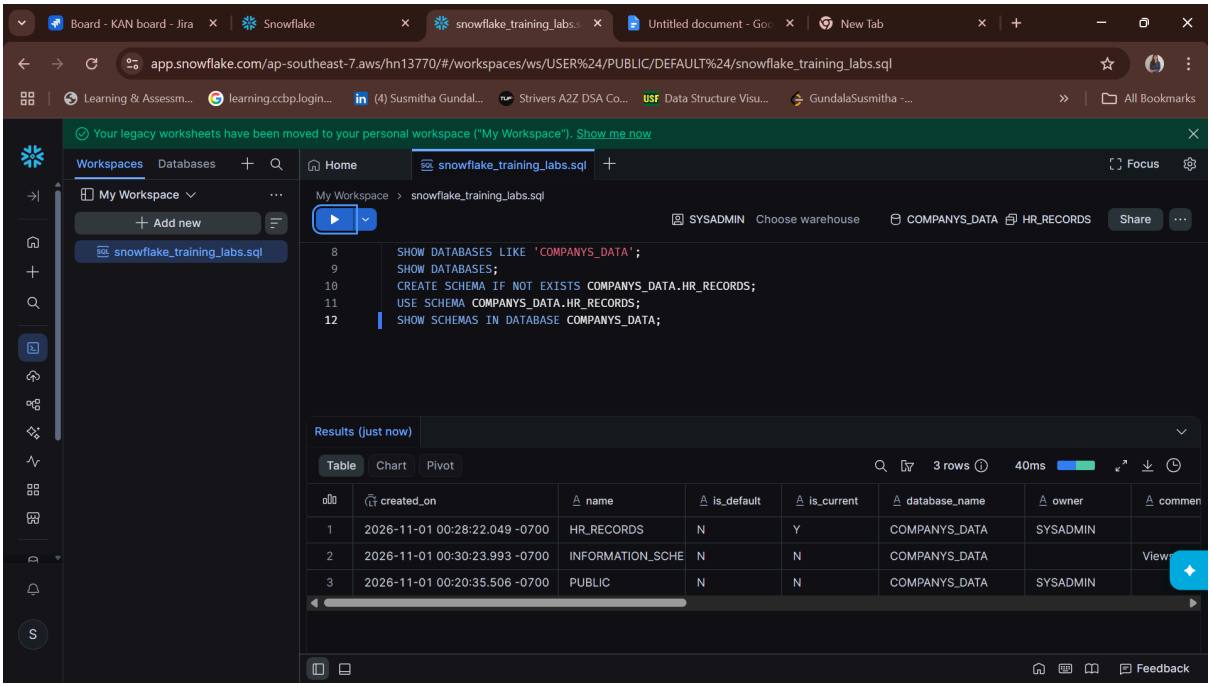
Select the schema:

```
SQL
USE SCHEMA COMPANYS_DATA.HR_RECORDS;
```



Verify:

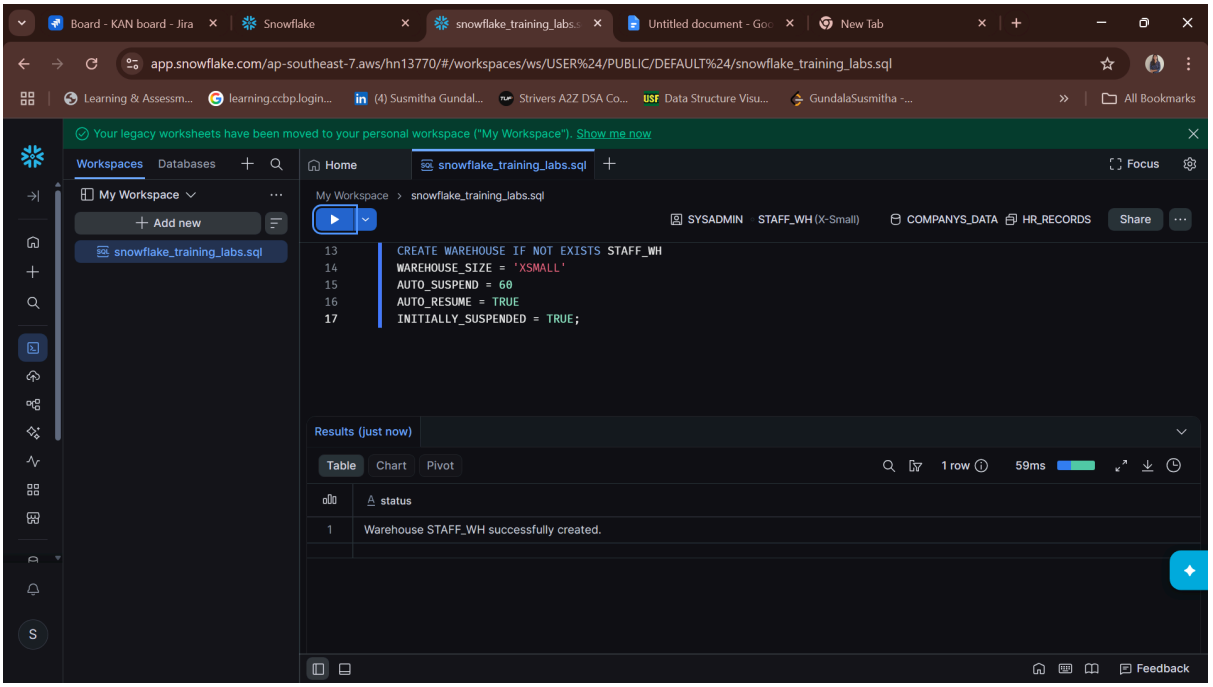
```sql
SHOW SCHEMAS IN DATABASE COMPANYS_DATA;
```



PART 2: VIRTUAL WAREHOUSE MANAGEMENT

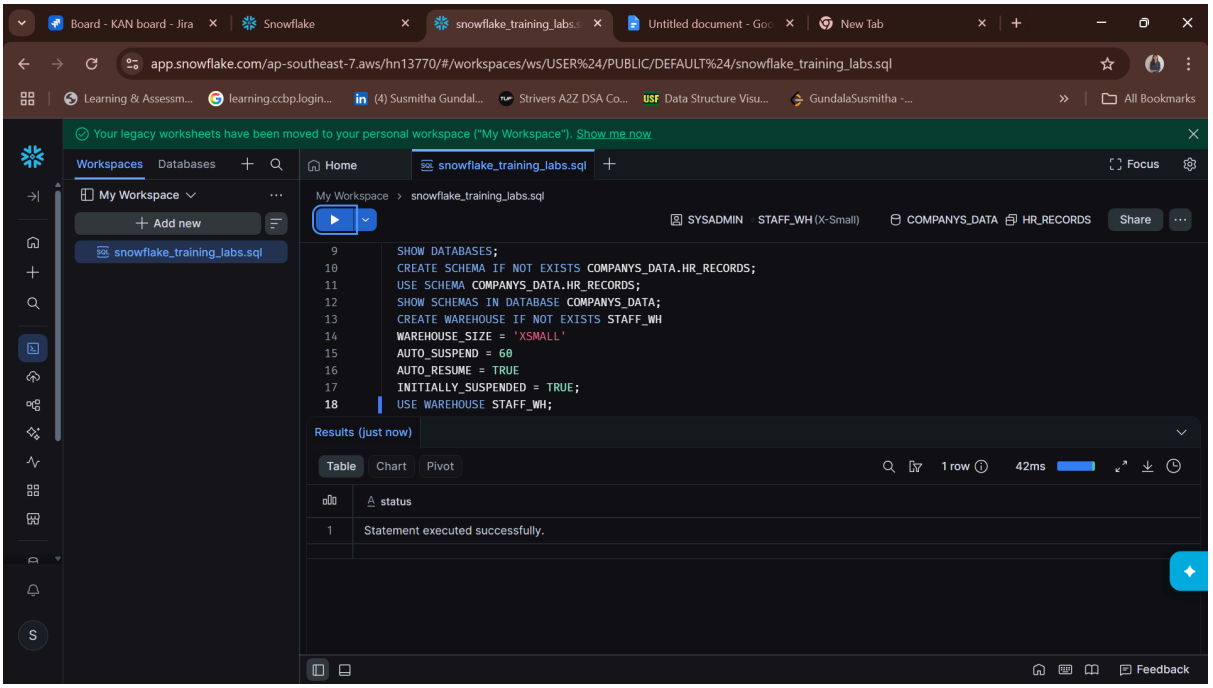
Step 4: Create a Virtual Warehouse

```sql
CREATE WAREHOUSE IF NOT EXISTS STAFF_WH
WAREHOUSE_SIZE = 'XSMALL'
AUTO_SUSPEND = 60
AUTO_RESUME = TRUE
INITIALLY_SUSPENDED = TRUE;
```



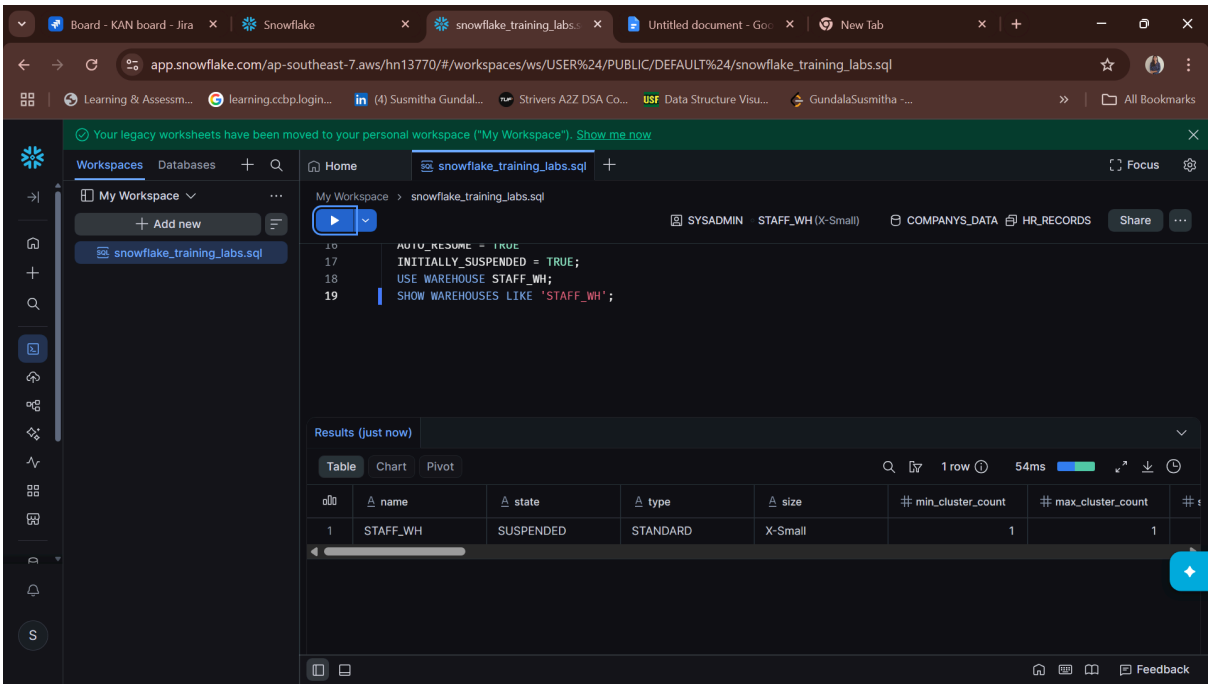
Select the warehouse:

```
SQL
USE WAREHOUSE STAFF_WH;
```



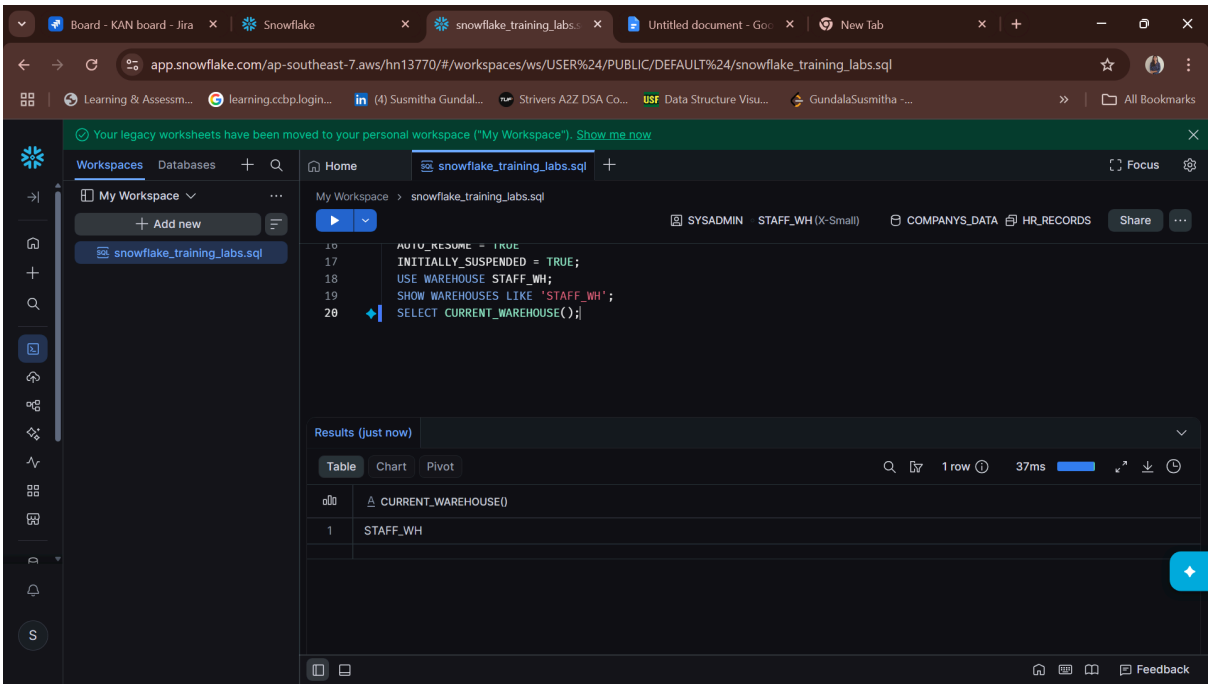
Verify:

```sql
SHOW WAREHOUSES LIKE 'STAFF_WH';
```



Check active warehouse:

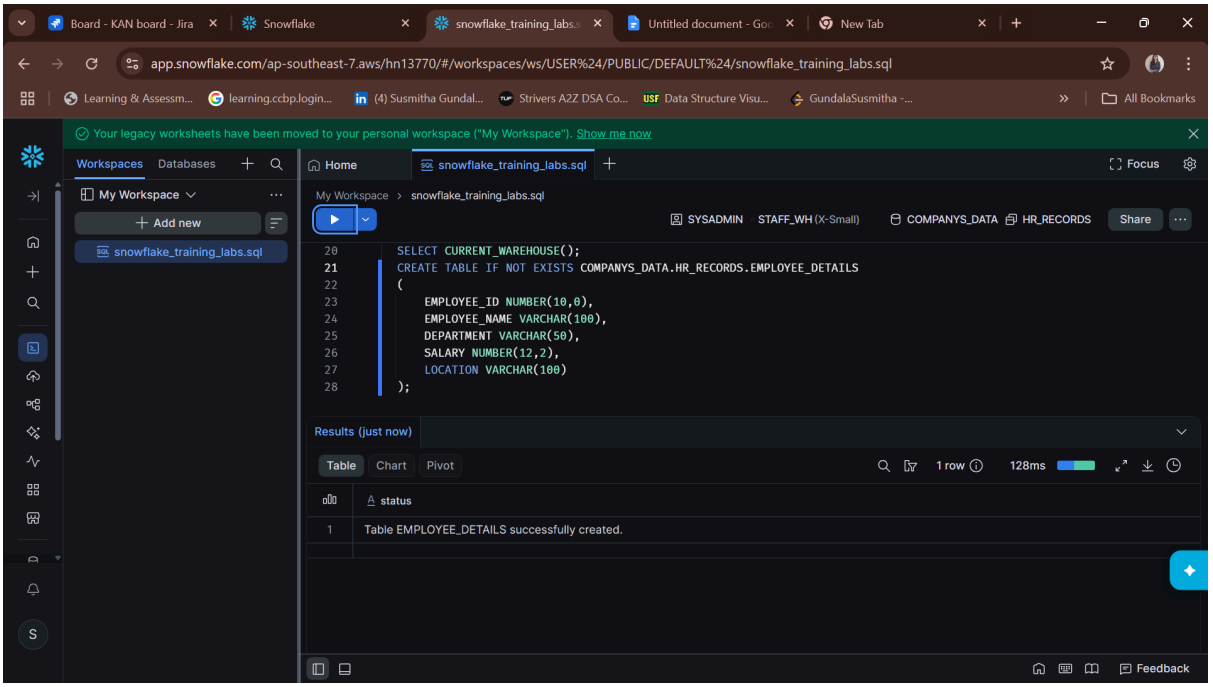
```sql
SELECT CURRENT_WAREHOUSE();
```



PART 3: TABLE CREATION

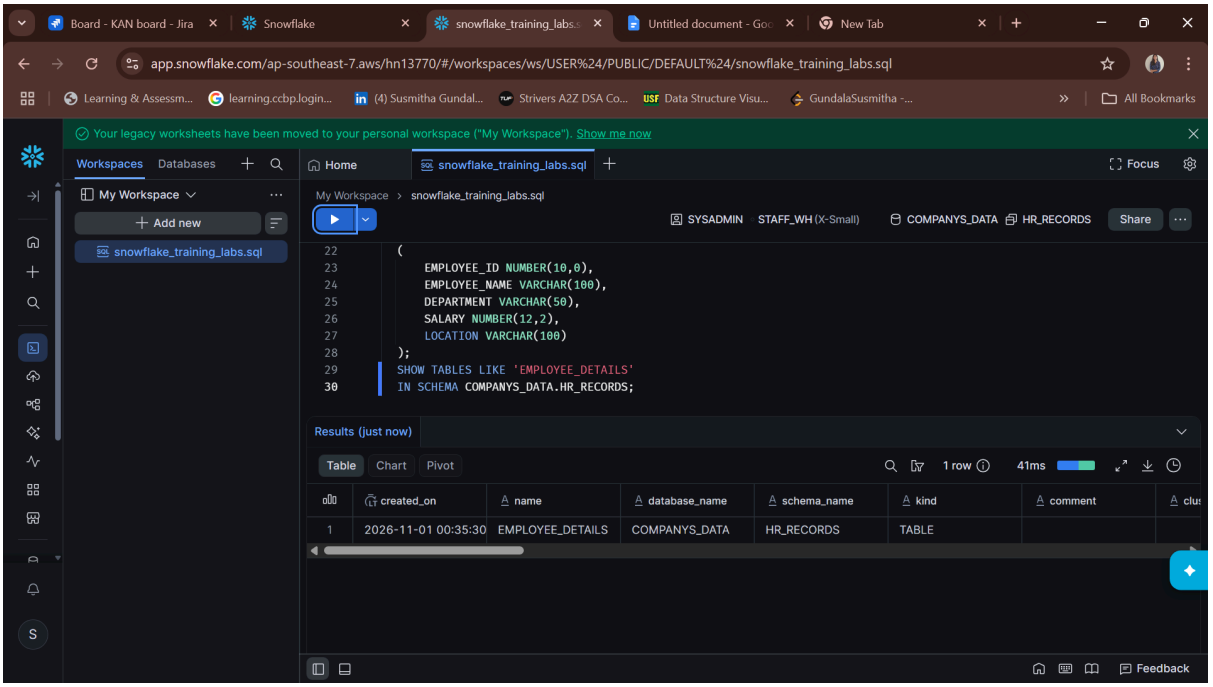
Step 5: Create an Employee Table

```sql
CREATE TABLE IF NOT EXISTS COMPANYS_DATA.HR_RECORDS.EMPLOYEE_DETAILS
(
    EMPLOYEE_ID NUMBER(10,0),
    EMPLOYEE_NAME VARCHAR(100),
    DEPARTMENT VARCHAR(50),
    SALARY NUMBER(12,2),
    LOCATION VARCHAR(100)
);
```



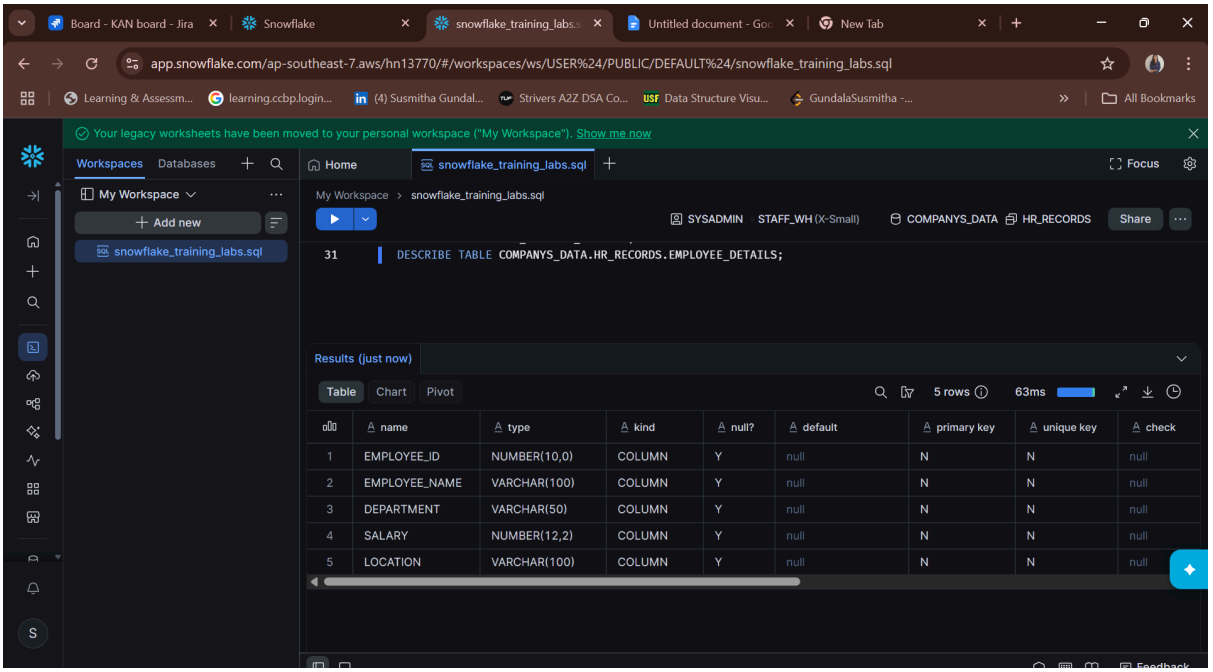
Verify table:

```sql
SHOW TABLES LIKE 'EMPLOYEE_DETAILS'
IN SCHEMA COMPANYS_DATA.HR_RECORDS;
```



Check table structure:

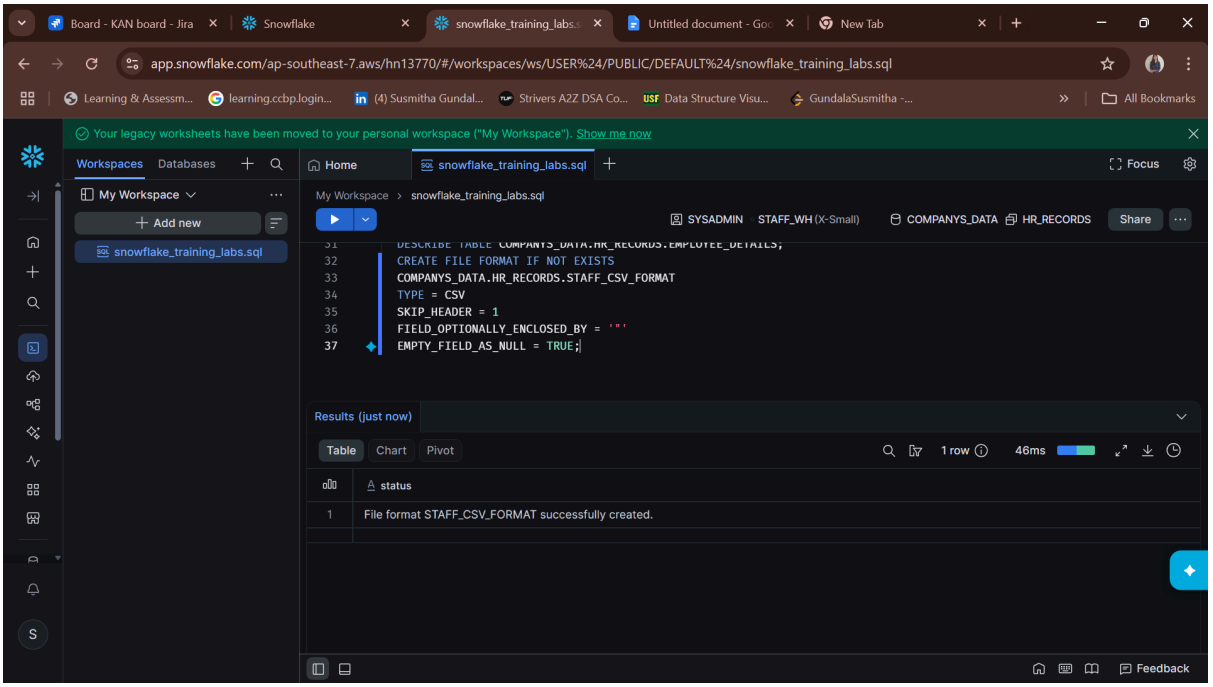
```sql
DESCRIBE TABLE COMPANYS_DATA.HR_RECORDS.EMPLOYEE_DETAILS;
```



PART 4: FILE FORMAT CREATION

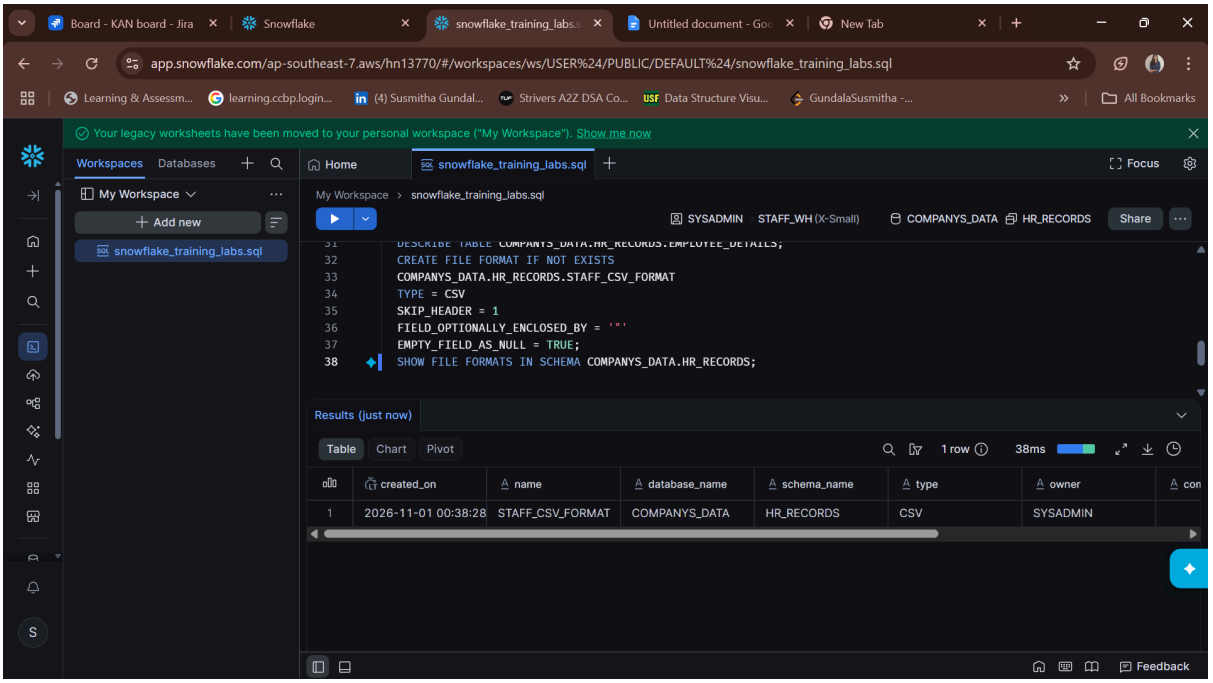
Step 6: Create a CSV File Format

```sql
CREATE FILE FORMAT IF NOT EXISTS
COMPANYS_DATA.HR_RECORDS.STAFF_CSV_FORMAT
TYPE = CSV
SKIP_HEADER = 1
FIELD_OPTIONALLY_ENCLOSED_BY = '"'
EMPTY_FIELD_AS_NULL = TRUE;
```



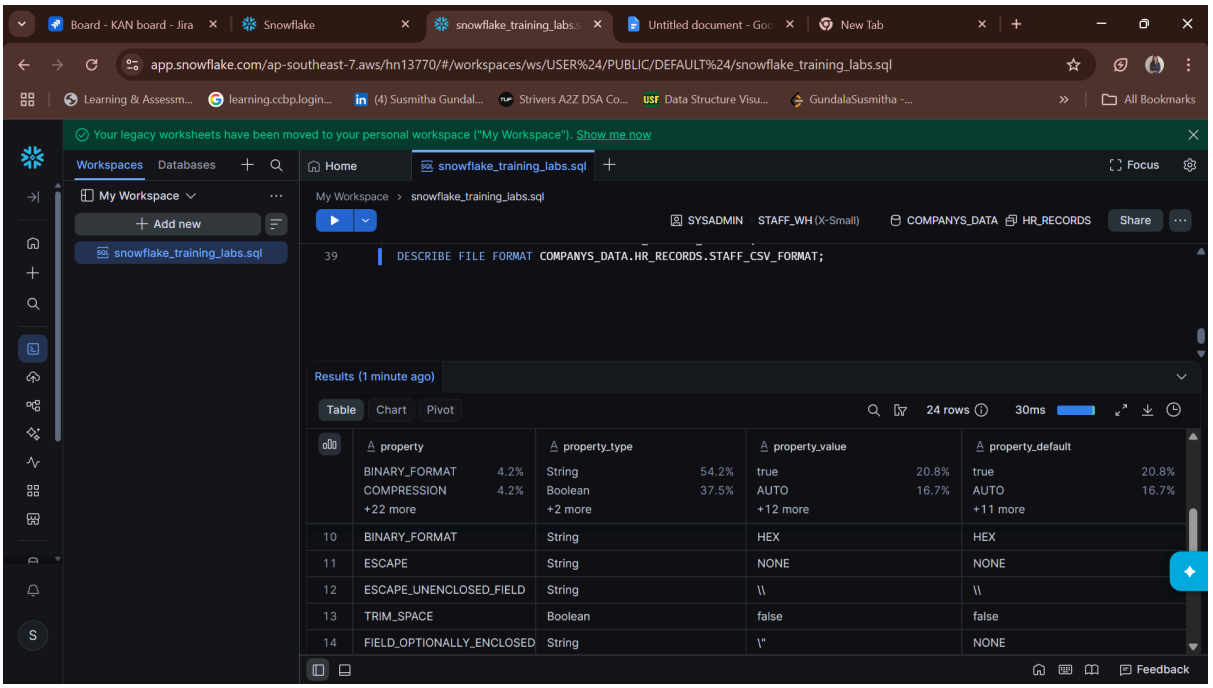
Verify:

```sql
SHOW FILE FORMATS IN SCHEMA COMPANYS_DATA.HR_RECORDS;
```



View file format properties:

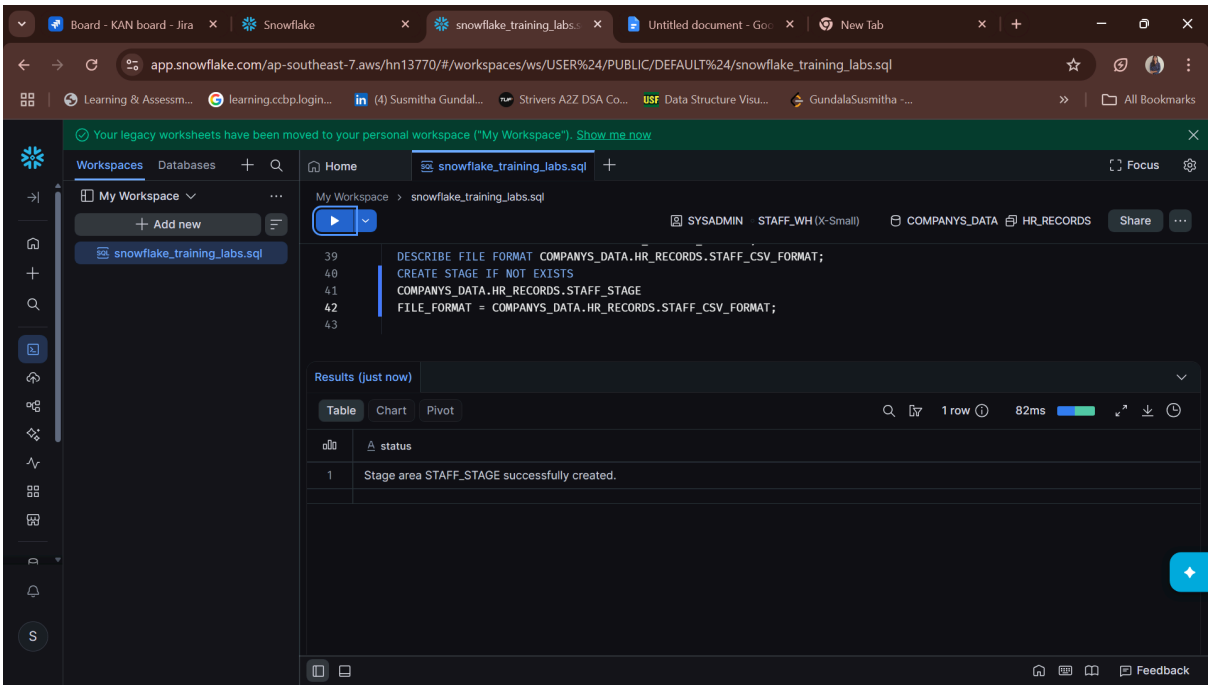
```
SQL
DESCRIBE FILE FORMAT COMPANYS_DATA.HR_RECORDS.STAFF_CSV_FORMAT;
```



PART 5: INTERNAL STAGE MANAGEMENT

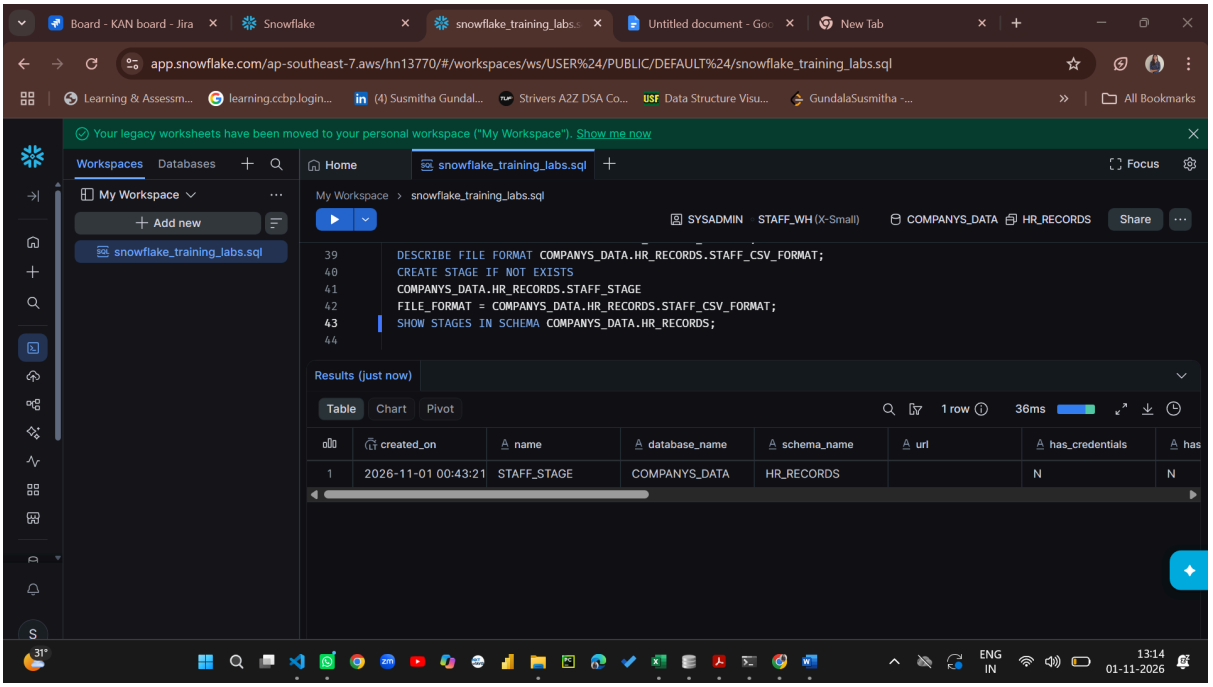
Step 7: Create an Internal Stage

```sql
CREATE STAGE IF NOT EXISTS
COMPANYS_DATA.HR_RECORDS.STAFF_STAGE
FILE_FORMAT = COMPANYS_DATA.HR_RECORDS.STAFF_CSV_FORMAT;
```



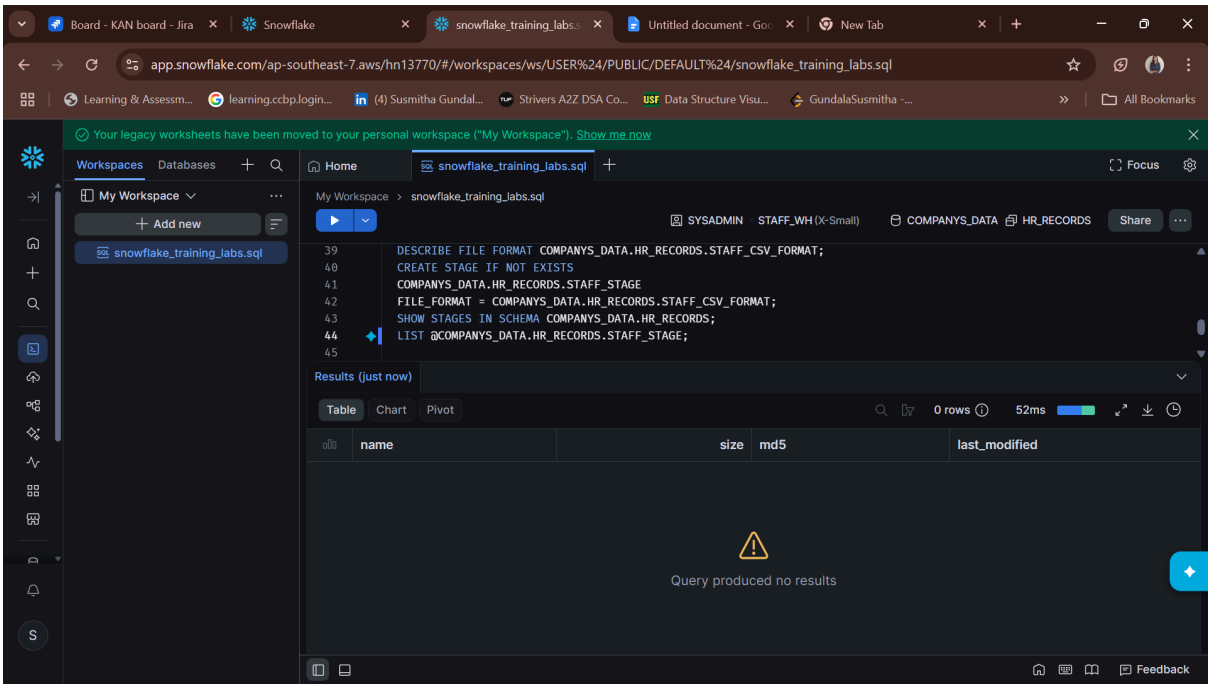
Verify:

```sql
SHOW STAGES IN SCHEMA COMPANYS_DATA.HR_RECORDS;
```



List the files inside the stage:

```
SQL
LIST @COMPANYS_DATA.HR_RECORDS.STAFF_STAGE;
```

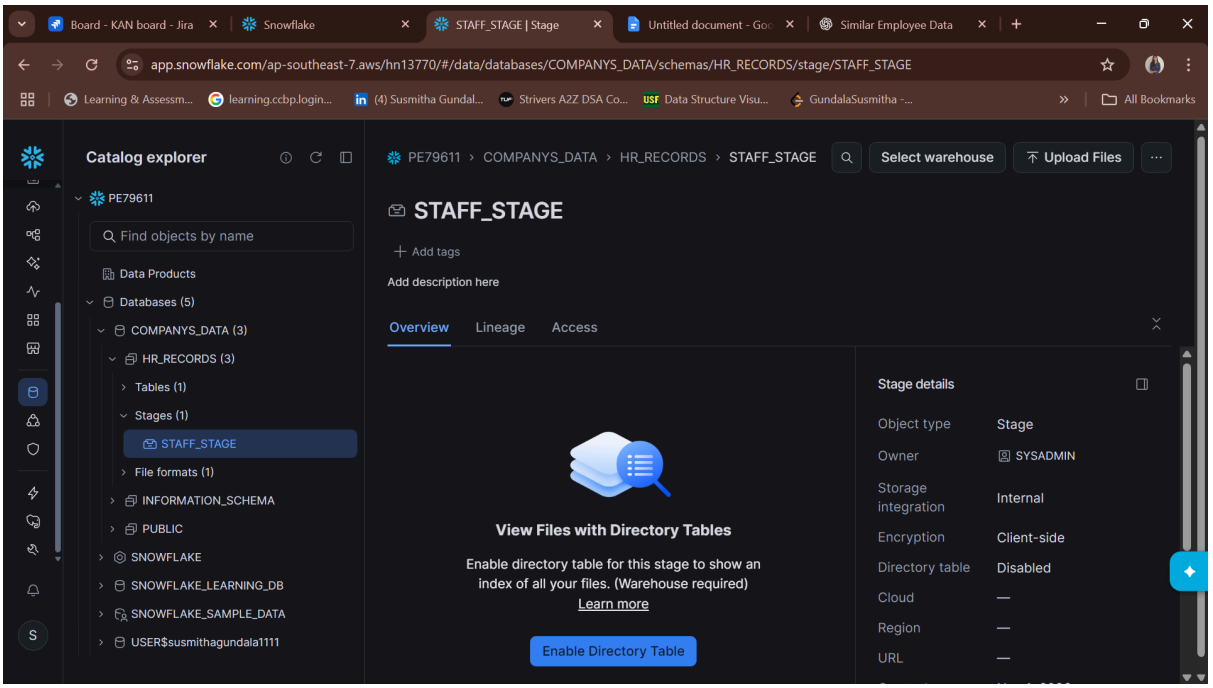


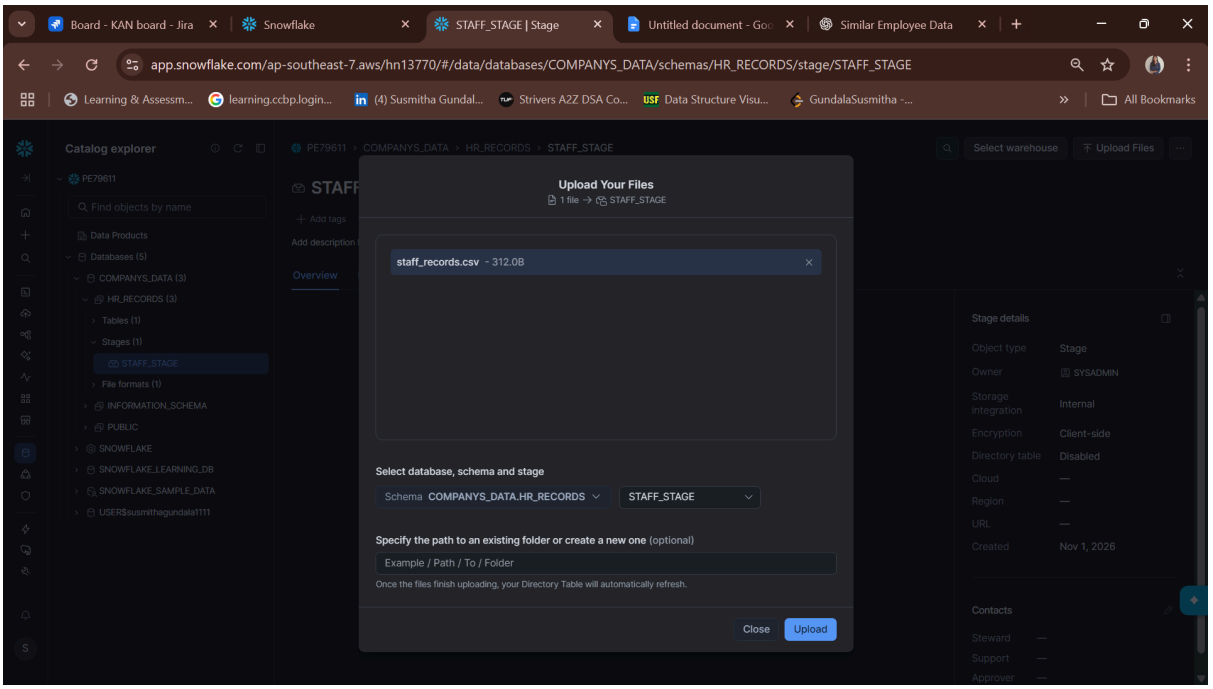
Step 8: Prepare the CSV File

staff_records.csv

None

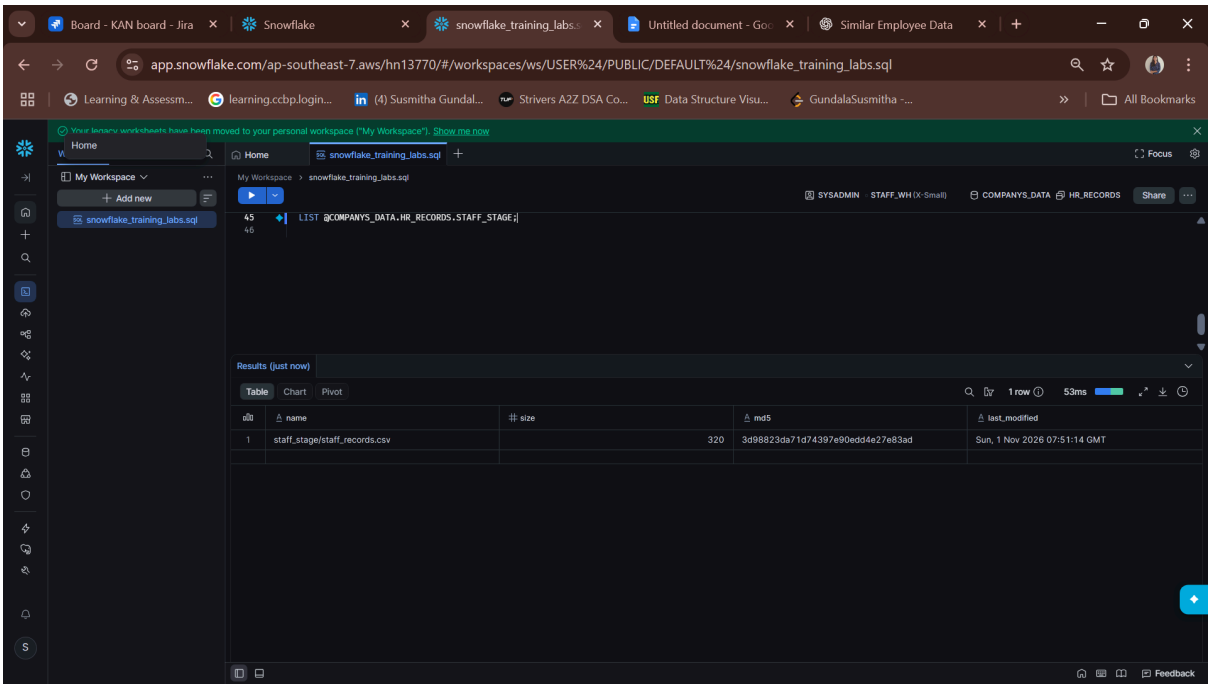
```
EMPLOYEE_ID,EMPLOYEE_NAME,DEPARTMENT,SALARY,LOCATION
3001,Arjun,IT,61000,Hyderabad
3002,Swathi,HR,48000,Chennai
3003,Vikram,Finance,72000,Bangalore
3004,Keerthi,IT,69000,Mumbai
3005,Manoj,Sales,45000,Chennai
3006,Lavanya,HR,52000,Bangalore
3007,Praveen,Finance,68000,Hyderabad
3008,Harsha,IT,64000,Mumbai
```





Verify the uploaded file:

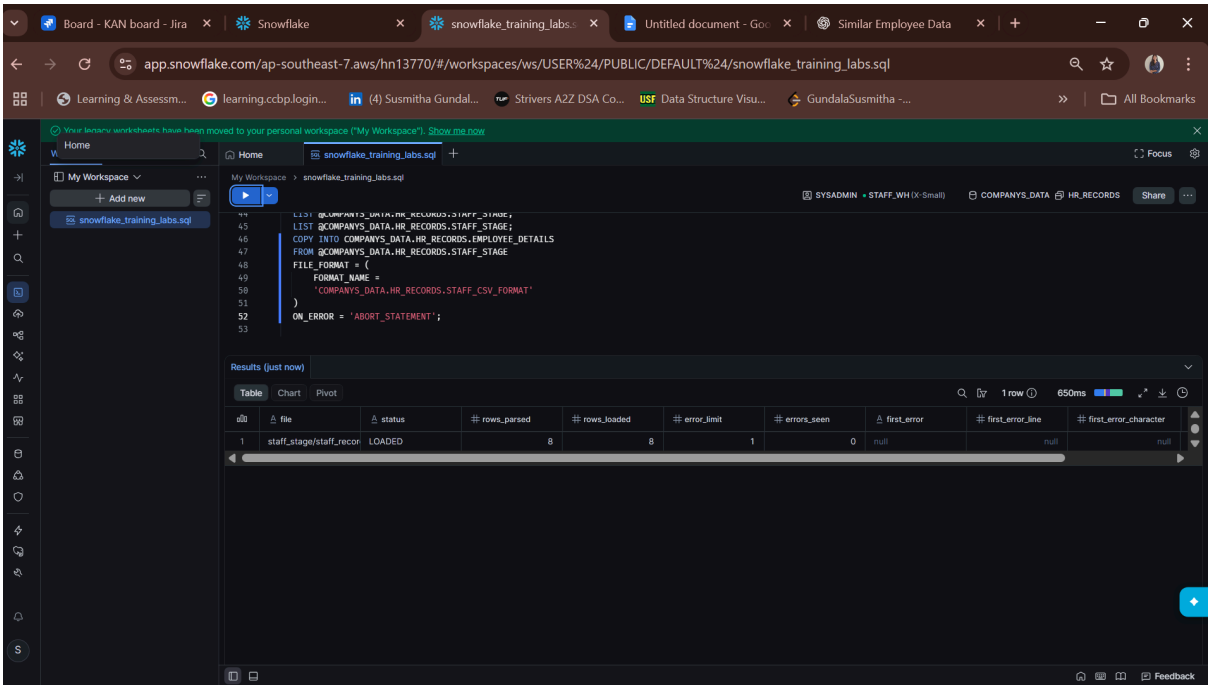
```sql
LIST @COMPANYS_DATA.HR_RECORDS.STAFF_STAGE;
```



PART 6: LOADING DATA INTO SNOWFLAKE

Step 9: Load CSV Data Using COPY INTO

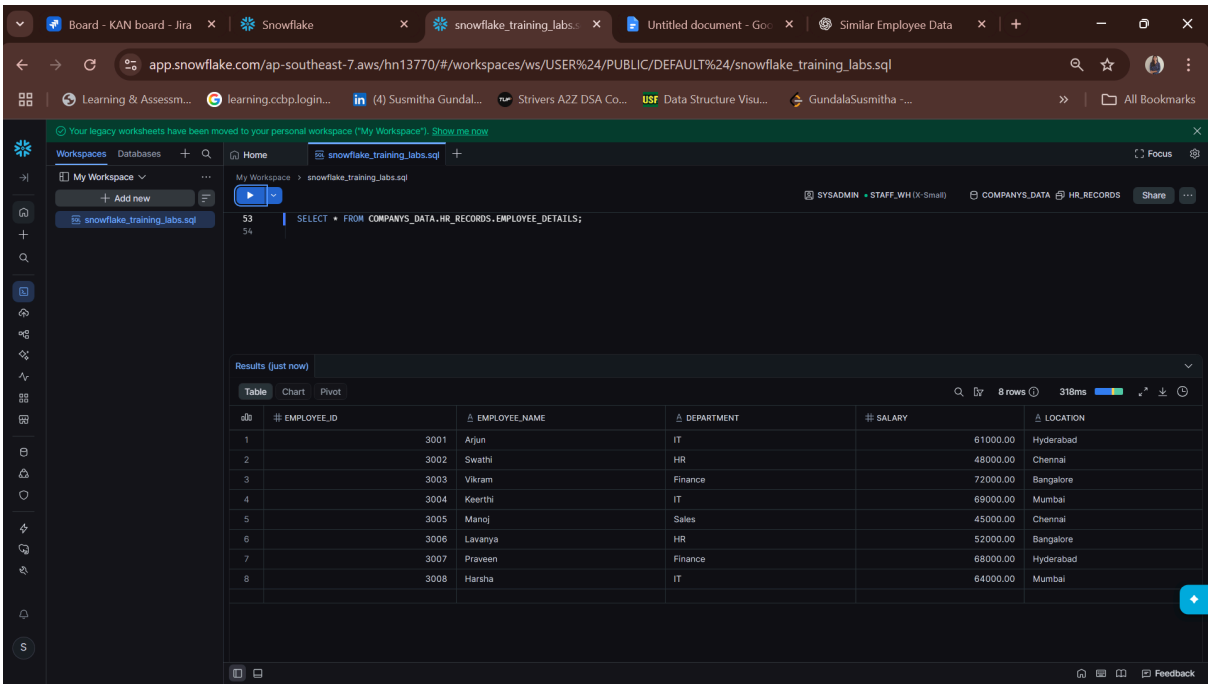
```sql
COPY INTO COMPANYS_DATA.HR_RECORDS.EMPLOYEE_DETAILS
FROM @COMPANYS_DATA.HR_RECORDS.STAFF_STAGE
FILE_FORMAT = (
    FORMAT_NAME =
    'COMPANYS_DATA.HR_RECORDS.STAFF_CSV_FORMAT'
)
ON_ERROR = 'ABORT_STATEMENT';
```



Step 10: Verify Loaded Records

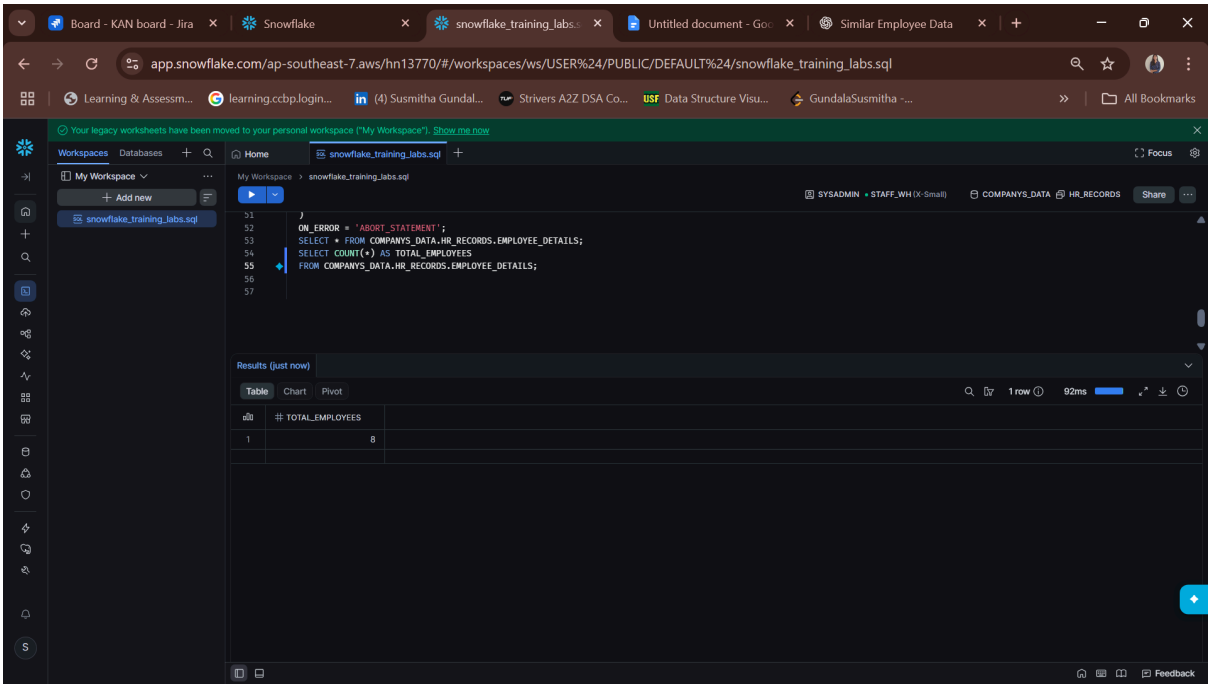
Display all records:

```sql
SELECT * FROM COMPANYS_DATA.HR_RECORDS.EMPLOYEE_DETAILS;
```



Count the records:

```sql
SELECT COUNT(*) AS TOTAL_EMPLOYEES
FROM COMPANYS_DATA.HR_RECORDS.EMPLOYEE_DETAILS;
```



Display the first five records:

```sql
SELECT *
FROM COMPANYS_DATA.HR_RECORDS.EMPLOYEE_DETAILS
LIMIT 5;
```

